

SparkSQL 2 — Project Defense

Outlier detection & categorical encoding

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1 • Problem & approach

Flag anomalous rows in the wine-quality dataset:

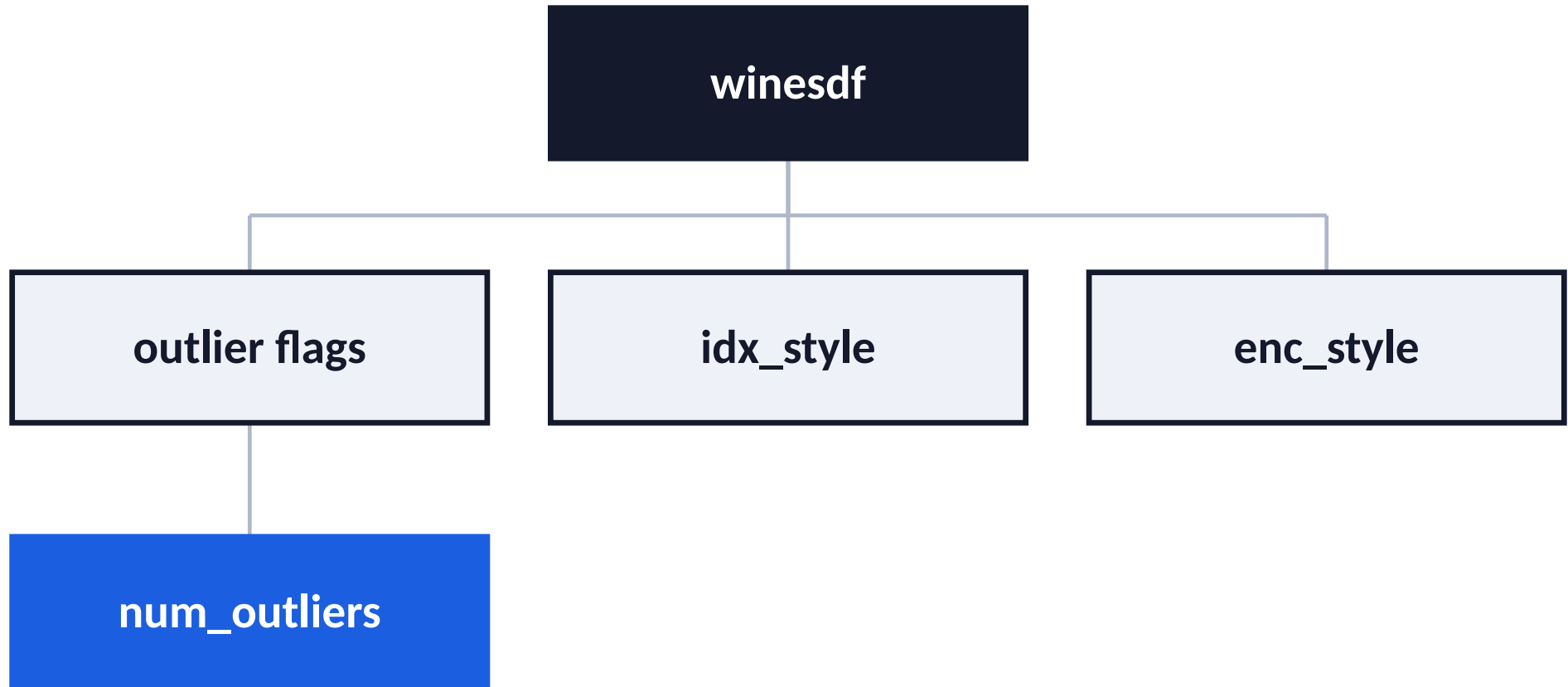
- per-variable outliers via the **IQR rule**
- keep rows anomalous in ≥ 3 dimensions

IQR rule

A value is an outlier if it falls below $Q_1 - 1.5 \cdot \text{IQR}$ or above $Q_3 + 1.5 \cdot \text{IQR}$, with $\text{IQR} = Q_3 - Q_1$.

Pipeline at a glance

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≈ 5%

of rows dropped as multi-dimensional outliers

Stage	Rows kept	Note
Raw	6 497	full dataset
After IQR filter	6 180	num_outliers < 3
Encoded	6 180	one-hot style

Enforced at the pipeline, not patched afterwards.

Timeline

2 · Results

